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Artificial Intelligence (AI) in the Automotive Industry and the use of Exoskeletons in the Manufacturing Sector of the Automotive Industry

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Abstract: **Introducing Artificial Intelligence (AI) in the Automotive Industry and developing related technologies will significantly impact the automotive industry. A technological innovation like Autonomous driving entails using artificial intelligence (AI), which represents the future of transportation and applications that will influence the concept of driving. New businesses related to mobility will emerge, and the already existing ones will have to adapt to the necessary shifts. Some security and AI algorithms are used in the vehicle security domain. This research aims to give some idea and understanding about how Artificial Intelligence (AI) is important and impacts the automotive industry. While discussing autonomous vehicles, there is a question to ask or focus on – what about the feasibility of autonomous production? Some light has been thrown on the benefits and requirements for automation of production plants, and its future has been discussed.**

Keywords – Artificial Intelligence, AI, Automobiles, Exoskeletons, AI in manufacturing, Exoskeletons in Manufacturing of autonomous vehicles.

I. INTRODUCTION

The automotive sector has been at the forefront of incorporating AI into its operations. AI has the potential to improve car safety, enhance the driving experience, and boost manufacturing efficiency. Artificial intelligence (AI) utilizes algorithms & data to duplicate human thinking and have decision-making ability. Here are some examples of how artificial intelligence is being applied in the automotive industry:

Autonomous Vehicles: Artificial intelligence has enabled the creation of self-driving automobiles, which are set to transform the automotive sector. Self-driving cars have the potential to improve road safety by lowering the number of accidents caused by human error. They can also help to minimize traffic congestion and fuel efficiency.

Predictive Maintenance: Artificial intelligence systems can examine vehicle data to forecast when maintenance is required. This can help to prevent breakdowns and extend vehicle lifespan.

Driving Assistance: Artificial intelligence (AI) can be used to create driver assistance systems that can help drivers avoid accidents by recognizing impediments and advising the driver. These technologies can also assist drivers in remaining in their lane and keeping a safe distance from other vehicles.

Manufacturing: AI is being utilized to improve efficiency and lower costs in the manufacturing process. Robots driven by AI, for example, may do repetitive activities like welding and painting, freeing up human workers to focus on more complicated tasks.

AI is an algorithm that assists the system in learning and solve problems independently. AI performs tasks and responds like humans do it intelligently and independently. It continues to learn from all the past experiences and keeps on adjusting to newer participations. It is being deployed and effectively used across industries like Information and Communication Technology (ICT), Automotive, Healthcare, financial services, Industrial Manufacturing, Life Science, Retail, Chemicals and Oil and gas. Financial services & IT are considered more sophisticated industries with respect to using AI for daily activities. In ICT, organizations offer AI-based assistants for commercial applications (examples are Alexa, Google Assistant and Siri). The automotive industry can be categorized as new adopters of AI technologies in terms of technology adoption and implementation of AI across aspects of the automotive industry.

Automation, technologies and data exchange are considered as trends of the industrial revolution. In present scenario, the industries are facing new challenging situations with respect to the competitiveness and demand in the market, and there is a need to undergo a radical change to evolve to Industry 4.0. AI enables improved decision-making dynamics. It helps make a precise decision for industry 4.0, which results in improved business performances, improved quality control, reduced machine failure, increased productivity, and lower costs.

Artificial Intelligence has several benefits that include -

- To use data to automate the learning process,
- To enhance the intellectual abilities of existing products,
- To do programming on the data available, there is need to adapt to intelligent learning algorithms,
- To analyze the rationality of data, and
- To Improve data accuracy.

AI indeed transforms the present world, but has its own constraints. The substantial challenges of AI include to learn from experiences, and there is no way that the knowledge can be

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incorporated into the learning. Apart from that, it becomes extremely challenging, when any discrepancy in the data is reflected in the results. Exoskeletons are wearable technologies that can improve humans' physical capabilities. Exoskeletons are being employed in the automobile industry to alleviate physical strain on personnel doing repeated jobs like lifting heavy goods. Exoskeletons can aid in preventing injuries and reducing worker weariness, resulting in greater production and efficiency. Following are some examples of how exoskeletons are employed in the automotive industry: Exoskeletons can help workers reduce physical strain by supporting their body weight and lowering the burden on their joints. This can aid in preventing injuries and reducing worker tiredness, resulting in greater production and efficiency. Exoskeletons can improve workstation ergonomics by providing support for workers who are standing or working in uncomfortable situations. This can assist to lower the risk of musculoskeletal problems and other workplace injuries. Exoskeletons can also improve worker performance by offering increased strength and dexterity. This is especially important when using tools or assembling components requiring accuracy and control. Enhancing workplace safety: Exoskeletons can also increase workplace safety by lowering the chance of accidents and injuries. Exoskeletons, for example, can be built to guard against falling items or sharp equipment. Overall, exoskeletons are a promising technology that can potentially improve the health and safety of automotive workers. As technology advances, we may anticipate increasingly inventive exoskeleton designs customised to the specific needs of automobile manufacturing processes. Fig.1. shows the industry based on the maturity level for AI Adoption.

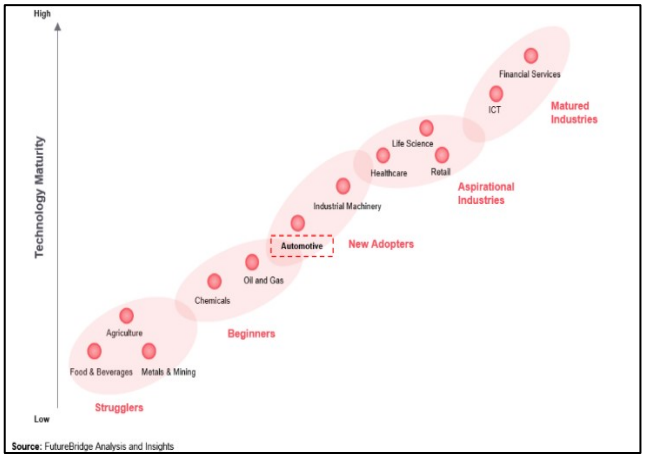

Fig.1 Industry based on the Maturity Level for AI Adoption

The article is organized as follows. Chapter-2 explains about the literature review. Chapter-3 explains about the research methodology, findings and applications. Chapter-4 describes the role of AI in the automotive industry. Chapter-5 concludes the article.

II. Literature Review

Bahman Javadi, Hamid Khayyam, Reza N Jazar and Mahdi Jalili from AI and IoT for autonomous vehicles talk about what artificial intelligence is and mentions that combinations and technologies like AI and IoT will transform the industry towards fully Autonomous vehicles. The effect of AI in the locomotive manufacturing states the impact & advantages of AI in the automotive sector and also applying driverless cars technology.

The papers by Hagemann, S., Sünnetcioglu, A., & Stark, R. (2019) and Gamer, Thomas, et al. (2020) mentions the already existing and established standards with six levels of autonomous driving. It also introduces the levels of autonomy and some key autonomous features of process operations. It also includes the efforts spent on production system design.

A survey of new orientations in vehicular cybersecurity applying artificial intelligence-based methods talks about vehicle cybersecurity. The research paper from Perez Luque, Estela, et al. (2020) talk about the Exoskeletons and its use in Manual tasks. However, there is minimal research about the application and use cases of Exoskeletons in the Automotive industry. This paper also tries to shed some light on the Exoskeletons used in manufacturing automobiles.

Vladimir Franki .et.al in [4] reviewed the role of AI in the power sector. They explored about the application of digital solutions in various sectors. The article also reviewed the different AI approaches in the power sector in detail.

The authors of article [5] examined previous CMS research and identified five critical enabling technologies for smart manufacturing in the automobile sector. The following topics are picked and described: digital twins, additive manufacturing, AI-based monitoring and inspection, human-robot cooperation, enhanced supply chain and logistics, and their current state and applications. In addition, a five-level framework for an automobile smart factory (SF) was also presented based on the basic keywords from each technology's trend.

The writers of article [6] explored the enormous monetization possibilities of data analytics and the use of artificial intelligence and blockchain throughout the automobile industry's product value chain and life cycle. The authors observed that the whole value chain and product life cycle could not be implemented in a single step. This was followed by other issues such as the required data availability, data cleansing/preparation, a different data owner, and a shift in staff culture.

Gueltoum Bendiab. et.al explored the implementation of Blockchain and Artificial Intelligence in article [7] in the security of Autonomous vehicles. The article combined the usage of Blockchain and AI in protecting AV against malicious vehicles. The authors also explored the combination of Blockchain, and AI solutions could be used to secure AVs. The authors also introduced the security and privacy threats that may arise in the application of AI for securing the Autonomous vehicles.

In article [8], the authors explored the impact of 4th industrial revolution on the transitory stage of the AI. They explored the new levels of organization and digitalization, focussed on the controlling monitoring the product life cycle. The article ensured that the digital transformation and sustainable development of the automobile industry, innovative production and communication systems, industrial and environmental regulations, and legislation established by automotive industries and suppliers keep them ahead of their competition.

In article [9], the authors highlighted about road markings, intersections for the role of a AV operation. This work aimed to describe how Autonomous Vehicles' spread will influence contemporary society in regard to urban transport systems. A comprehensive analysis of the expected developments within urban transport systems was presented, and some crucial issues concerning benefits and drawbacks were also discussed in this article.

The article [10] writers identified the impact of AI in the automobile sector. The writers utilised a descriptive research approach to collect data from known facts. The writers concluded that Industry 4.0 demands the supply of amenities that humans

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require now. The existence of Artificial Intelligence technology, which is intended to address human requirements with automation technology in all sectors of life support, includes the automotive/transportation industry.

III. Research Methodology

The research done is qualitative in nature. This research paper includes secondary data as a part of the research. There are references taken from research papers and the internet. The references used for this research paper are authentic in nature and the objective of the sources are reliable in nature. The research is focused on finding the uses and applications of Artificial intelligence in the Automotive industry across various sectors. Some keywords used to find relevant papers and data include AI in the automotive sector, Exoskeletons in the automotive industry, etc. By skimming through research papers, it is understood how AI is used in manufacturing, transportation, aftermarket services, and so on. Focus is given to how AI is used in the manufacturing of automobiles and the use of exoskeletons in the manufacturing process. The study period is around last 4 to 5 years as advancements regarding AI have been witnessed in this duration.

3.1 Findings and Applications

We are on the verge of entering industry 4.0. AI has been developed and would enable the computers to do predictions of results based on data given as input; which is known as "Deep Learning," where the technology enables the computers take decisions on their own to perform those decisions by the other components swiftly. This technology integrates algorithms with artificial neural networks that will enable computers to grasp abstract and complex concepts. By incorporating technology components in the industrial era, 4.0 can produce an autonomous car that can steer the freeway with a determined scenario. AI has been implemented on driverless cars by using intelligent parking assist and crash-avoidance systems. When an autonomous car is turned on, the commuter should enter information like the destination data to enable the computer to make the decision automatically. All the data and insights generated from autonomous cars, which includes radar sensors, the camera distance from nearby objects such as traffic signals, pedestrian sensors, curbs, and markings on the road, will also affect the car's speed.

IV. Role of Artificial Intelligence in the Automotive Industry

Implementation of AI in the manufacturing of automotive includes design, supply chain, production, and post-production. Implementation of AI is also done in 'driver assistance' and 'driver risk assessment' systems. These implementations of AI are transforming the transportation sector along with driver assist, autonomous driving, and cloud computing. The aftermarket services like predictive maintenance, servicing data and vehicle insurance are also revolutionizing with the utilization of AI.

4.1 AI in the manufacturing sector

Systems and robots are developed that learn automotive manufacturing skills and work simultaneously with humans in manufacturing automotive industry. All the activities like designing, parts production, and assembly are done with AI's assistance. AI is helping humans in manufacturing of automotive with the use of Exoskeletons. The nature of work done at the assembly lines of automotive manufacturing units often leads to injuries and musculoskeletal pain. In the United States, every year 10% of the workers suffer from work related musculoskeletal disorders (WMSD). The exoskeletons are used as support to body weight. They assist with lifting, help maintain loads, and even stabilize the worker's body. These exoskeleton systems assist the arms, upper-body, and lower-body. They tend to improve the ergonomics of manual operation in vehicle manufacturing. Some high-tech exoskeleton systems are being explored by automakers like Hyundai, Ford, and General Motors. It solves the problem of an aging workforce and a lack of skilled workers in the industry. It is researched that use of Exoskeletons can reduce the muscle effort of workers by 60% in automobile industry. The nature of the work performed at the assembly line in the automotive sector is repetitive and, consequently, often causes musculoskeletal pain and injuries. But these exoskeleton systems make up for these issues, and manufacturing does not experience any lost workdays or lost productivity and ensures smooth functioning. The concept of Exoskeleton was first introduced in 1966. There are very limited number of certified exoskeletons available in the market. Smart Exoskeletons are customizable according to the varying conditions and combinations of the worker's body. These exoskeletons have huge potential to empower the workers daily and simultaneously decrease the work-related health issues. The fig.2 shows the different models of exoskeletons.

Fig.2 Models of Exoskeletons

4.2 AI in Transportation and Logistics

AI has been used widely in modern vehicles and played a pivotal role in driver assistance technology. At present, Tesla has been the most prominent example proving the power of AI in cars. AI can also help in logistics processes like managing the warehouse, factories, and machines – their runtime. AI helps by gathering data and analysing it for predictive maintenance. It can help avoid breakdowns and failures.

4.3 AI in Aftermarket Services

Several AI applications anticipate the issues relating to battery performance, engine performance, etc., which have a possibility of occurring in the future. Over and above that, several AI-powered insurance companies provide swift services like availing insurance policies and settlements of claims. The insurance companies use AI in cars to make real time risk assessments. For insurance claims, AI-powered video app can guide users by giving instructions to record video of their damaged vehicle.

4.4 Limitations & Future Research Direction

After going through all the research and information, AI, along with emerging technologies, remains a promising field to watch out for in the near future. However, the extent of implementation and adoption of AI in the automotive industry is still a question across the globe. The predictions made about the role of AI in the automotive industry are certainly promising to watch out for. However, the economic and global conditions will decide its fate. Emergencies like the pandemic and economic tensions between countries may make it difficult for the automotive industry to fully implement all the uses of AI and other emerging technologies supported by AI. In the future, the research can be directed towards where privacy will head when AI collects so much

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data, including 24/7 tracking locations, voice sensing, etc. The field of vehicular cybersecurity can be further researched as applying artificial intelligence-based methods talks about vehicle cybersecurity. Currently, the hardware of the vehicle generates a maximum value for the vehicle, and the vehicle's remaining value is generated by its software & content. In coming years, service offerings that are AI-powered technology (software related) within automotive are anticipated to turn into the Unique Selling Proposition for the vehicles, and the value generated by these software related technologies will be about 50 % of the total vehicle value. This certainly indicates that the technology providers offering solutions based on AI will gain traction as we advance and dominate the industry. The automotive industry will witness many acquisitions, collaborations, and partnerships in the future. As automotive OEMs (Original Equipment Manufacturers) are lagging in implementing AI technologies in their vehicles as compared to the technology providers.

V. Conclusion

In the automotive sector, the implementation of AI is gaining momentum. In 2015, the vehicles having some or other form of AI system installed was only 5 to 10%. However, lately, AI has been deployed in level 1 and 2 autonomous vehicles, for parking assistance, cruise control which has led to increased adoption of this technology. By the end of 2030, AI is expected to be a component of many automotive elements and applications, and it is believed that 95 to 98% of the modern vehicles will have AI as prime technology. According to Future Bridge's estimates, autonomous vehicle sales accounted for 33 million units in 2019, and it is expected to reach 80 million units by 2030. The market is expected to witness a CAGR of 8–10% during 2019–2030. In the next 10 years, AI in the automotive industry, as a technology, is believed to reach the maturity stage. The predictions say that the Level 3 autonomous vehicles are believed to account for most of the market share. Next to follow will be level 2 and then Level 4/5 autonomous vehicles. However, the Original Equipment Manufacturers have evaluated the promising future of automotive and have started collaborating with tech organizations to develop autonomous vehicles in partnership. To summarise, artificial intelligence and exoskeletons are revolutionising the automotive sector by boosting safety, efficiency, and production. As technology advances, we should expect additional improvements in the automobile sector to make vehicles safer and more efficient and manufacturing procedures more streamlined.

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